COA Master Study Guide (Units 3, 4, 5, 6)

UNIT 3 – COMPUTER ARITHMETIC

Priority 1 (Must Do)
Floating Point Representation: IEEE 754 Format, Mantissa, Exponent, Sign Bit, Normalization
Floating Point Arithmetic: Addition, Subtraction, Normalization after operation

Priority 2
Integer Representation: Sign Magnitude, 1's Complement, 2's Complement
ALU: Functions, Block Diagram, Operations Performed

Priority 3
Binary Arithmetic: Binary Addition, Binary Subtraction
Subtraction Using Complements: 1's Complement Method, 2's Complement Method

UNIT 4 – MEMORY ORGANIZATION

Priority 1
Cache Memory: Need, Organization, Direct Mapping, Associative Mapping, Set Associative Mapping, Advantages
Virtual Memory: Concept, Paging, Address Translation, Advantages
RAID: RAID Levels, Advantages, Applications

Priority 2
SRAM vs DRAM: Working, Characteristics, Comparison
Error Detection: Parity Bit, Checksum, CRC
Types of Errors: Single Bit, Multiple Bit, Burst Error

Priority 3
Semiconductor Main Memory, Memory Hierarchy, Memory Controllers

UNIT 5 – CONTROL UNIT

Priority 1
Hardwired Control Unit: Architecture, Working, Diagram, Advantages, Disadvantages
Microprogrammed Control Unit: Architecture, Working, Diagram, Advantages, Disadvantages
Hardwired vs Microprogrammed Control Unit: Complete Comparison

Priority 2
Micro-Operations: Register Transfer, Arithmetic, Logic
Role of CU in Micro-Operations: Control Signals, Register Operations, ALU Operations

Priority 3
Instruction Fetch, Instruction Decode, Timing Signals, Control Signal Generation

UNIT 6 – INPUT/OUTPUT ORGANIZATION

Priority 1
DMA: Working, DMA Controller, Diagram, Advantages
Interrupt Driven I/O: Working, Diagram, Advantages
Programmed I/O vs Interrupt I/O vs DMA: Comparison Table
Instruction Pipelining: Concept, Pipeline Stages, Hazards, Advantages

Priority 2
I/O Module: Need, Working, Diagram
Functions of I/O Module: Control & Timing, Communication, Buffering, Error Detection
Cache Coherence: Concept, Need
MESI Protocol: Modified, Exclusive, Shared, Invalid

Priority 3
External Devices, I/O Channels and Processors, External Interface, Multiple Processor Organization, SMP

MOST PROBABLE EXAM QUESTIONS

UNIT 3

1. Explain Floating Point Representation (IEEE 754 Format).
2. Explain Floating Point Arithmetic with suitable example.
3. Explain ALU with block diagram.
4. Explain Integer Representation methods.
5. Perform subtraction using 1's Complement.
6. Perform subtraction using 2's Complement.

UNIT 4

1. Explain Cache Memory with suitable diagram.
2. Explain Cache Mapping Techniques.
3. Explain Virtual Memory with diagram.
4. Explain RAID and RAID Levels.
5. Differentiate SRAM and DRAM.
6. Explain Error Detection Techniques.
7. Explain CRC.

UNIT 5

1. Explain Hardwired Control Unit with diagram.
2. Explain Microprogrammed Control Unit with diagram.
3. Differentiate Hardwired and Microprogrammed Control Unit.
4. Explain Micro-Operations with examples.

UNIT 6

1. Explain DMA with block diagram.
2. Explain Interrupt Driven I/O.
3. Compare Programmed I/O, Interrupt Driven I/O and DMA.
4. Explain Instruction Pipelining.
5. Explain I/O Module and its functions.
6. Explain MESI Protocol.

TOP 10 QUESTIONS (DO NOT SKIP)

1. Cache Memory
2. Cache Mapping Techniques
3. Virtual Memory
4. RAID and RAID Levels
5. DMA
6. Programmed I/O vs Interrupt I/O vs DMA
7. Instruction Pipelining
8. Hardwired Control Unit
9. Microprogrammed Control Unit
10. Hardwired vs Microprogrammed Control Unit